Supplementary Note 2: Comparing the fit of the Langevin model against a matched-complexity Gompertz hazard

Motivation

A reviewer asked for a quantitative comparison of our state-dependent Langevin model against an explicitly age-dependent alternative: *"...The authors should provide a quantitative comparison: at minimum, a statistical test asking whether an age-dependent model describes the empirical survival data significantly better than a Markovian one."* We address this directly: we score both model families against the real grouped survival data using an identical likelihood construction, and compare them by Akaike Information Criterion (AIC), which penalizes each model for its own number of fitted parameters.

Models compared

Our model is the Langevin process used throughout the main text ($dz/dt = \alpha z + gz^2 + \sqrt{2D}\,\eta(t)$, death at first passage to $Z = \alpha/g$). Its parameters are those already fit in the main text (not re-fit here): (α, g, D) for DMSO_day_10, and the post-switch (α_1, Z_1) for Auxin_day_21, with phase 1 fixed at the DMSO_day_10 fit.

The alternative is the classical Gompertz hazard, $M(t) = M_0 e^{\alpha_g t}$ – an explicit function of calendar age with no state variable – fit fresh to each condition by maximum likelihood, with the same parameter count as the corresponding Langevin model. For Auxin_day_21, phase 1 is likewise fixed (at the Gompertz model's own DMSO_day_10 fit) so that only the post-switch $(M_{0,2}, \alpha_{g,2})$ are estimated from that condition's data – mirroring exactly how the Langevin model only re-fits its own post-switch parameters at the intervention.

Both models are scored with the same discrete-time, grouped-data log-likelihood: each observed death at age t_i contributes the model's interval probability mass $S(t_{i-1}) - S(t_i)$; each censored worm at t_i contributes $S(t_i)$. The two families differ only in how $S(t)$ is produced – simulated (Langevin) or closed-form (Gompertz) – so neither model is scored by an easier rule than the other.

Results

Condition	Model	k	$\log L$	AIC
DMSO_day_10	Langevin (this work)	3	-408.1	822.1
DMSO_day_10	Gompertz	2	-342.9	689.8
Auxin_day_21	Langevin (this work)	2	-457.6	919.1
Auxin_day_21	Gompertz	2	-438.7	881.4

Table 1: Grouped-data maximum-likelihood fit of our Langevin model (parameters from `0_auto_fit_parameters_mle.py`, not refit here) versus a matched-complexity Gompertz hazard, per condition. Phase 1 is fixed at the DMSO_day_10 fit in both Auxin_day_21 models, so only phase-2 parameters ($k = 2$ each) are estimated from that condition's own data. $\Delta\text{AIC} = \text{AIC}_{\text{Gompertz}} - \text{AIC}_{\text{Langevin}} = -132.3$ (DMSO_day_10), -37.7 (Auxin_day_21); negative values favor Gompertz.

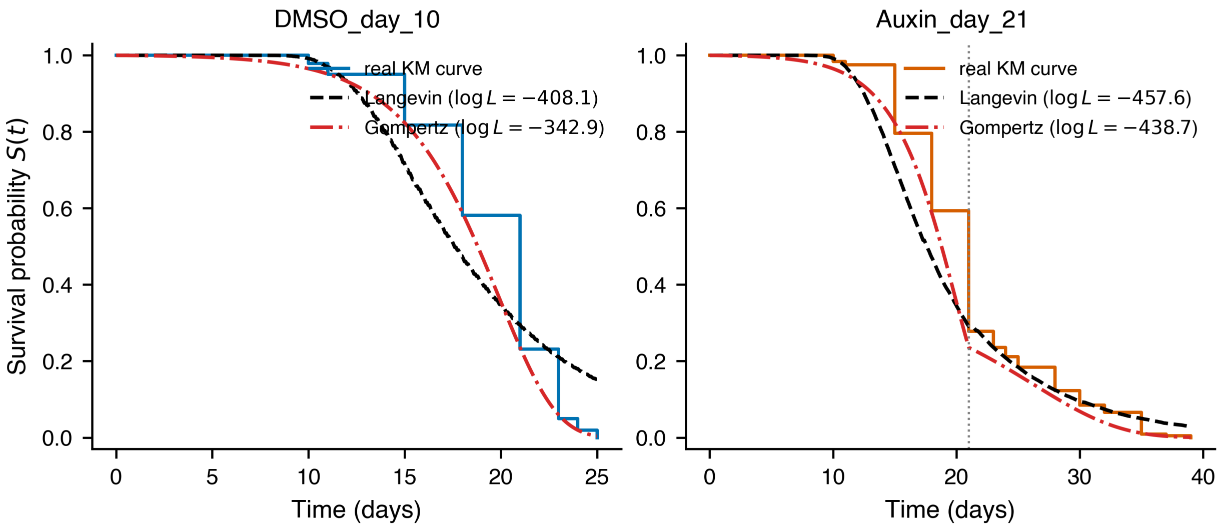

Figure 1: Real Kaplan–Meier survival curves (solid steps) for DMSO_day_10 (left) and Auxin_day_21 (right), overlaid with the fitted Langevin model (dashed) and fitted Gompertz model (dash-dotted). The vertical dotted line marks the day-21 intervention.

Table 1 reports the log-likelihood, parameter count, and AIC for each model and condition. For DMSO_day_10, the Gompertz hazard achieves a substantially higher log-likelihood despite using one fewer parameter, and the resulting ΔAIC favors it decisively by conventional thresholds ($|\Delta\text{AIC}| > 10$). For Auxin_day_21, with phase 1 fixed identically in both models so that the comparison isolates the post-switch hazard shape, the Gompertz hazard again achieves the higher log-likelihood at matched parameter count, and again wins decisively on AIC.

Interpretation

By this test, on the currently available grouped survival data, an explicitly age-dependent Gompertz hazard describes both conditions at least as well as – and by AIC, better than – our state-dependent Langevin model at matched or lower parameter count. This result is specific to the population-level hazard-shape comparison that grouped survival data can support: neither model's fit quality bears on whether an individual worm's future dynamics depend only on its own current state z, since z is never directly measured in this dataset. We report this outcome plainly, per the reviewer's own suggested fallback, so that the manuscript's claim can be moderated accordingly rather than resting on visual inspection of simulated trajectories alone.